

PROBLEM

CASE STUDY 1

SMART MEDICAL DIAGNOSIS SYSTEM

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system uses propositional logic rules based on symptoms such as fever, cough, rash, and breathlessness to determine possible diseases such as Flu, Measles, and Pneumonia.

CASE STUDY 2

AUTONOMOUS TRAFFIC MANAGEMENT AGENT

A smart city traffic controller uses a First-Order Logic (FOL) knowledge base to manage emergency vehicle routing. The system uses information about vehicles, emergency types, clear paths, green signals, red signals, and vehicle locations to decide whether an emergency vehicle should proceed.

CASE STUDY 3

AGRICULTURAL EXPERT REASONING SYSTEM

An agricultural advisory agent uses a propositional knowledge base to recommend suitable actions for crops. Based on facts such as SoilDry and CropWheat, the system determines whether to ApplyDripMethod and whether there is a CropAtRisk.

CASE STUDY 4

WUMPUS WORLD KNOWLEDGE AGENT

An AI agent operates in a 4×4 Wumpus World environment using percepts such as Stench, Breeze, and Glitter. The agent applies propositional logic and inference rules to determine the possible locations of the Wumpus and Pit and to evaluate whether a particular cell is safe.